FUNDAMENTALS OF DATA MINING

Lecture 6

Clustering

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This Lecture…

- Hierarchical Clustering
  - Agglomerative Clustering

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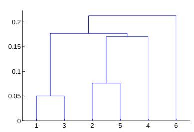
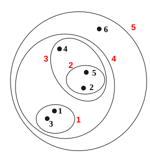
Hierarchical Clustering

- Produces a set of nested clusters organized as a hierarchical tree
- Can be visualized as a dendrogram
  - A tree like diagram that records the sequences of merges or splits





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Strengths of Hierarchical Clustering

- Do not have to assume any particular number of clusters
  - Any desired number of clusters can be obtained by 'cutting' the dendogram at the proper level
- They may correspond to meaningful taxonomies
  - Example in biological sciences (e.g., animal kingdom, …)

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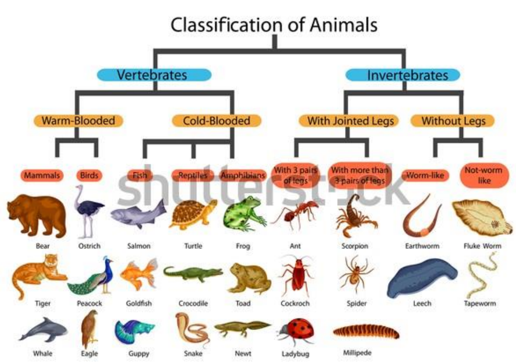

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Hierarchical Clustering

- Two main types of hierarchical clustering
  - Agglomerative:
    - Start with the points as individual clusters
    - At each step, merge the closest pair of clusters until only one cluster (or k clusters) left
  - Divisive:
    - Start with one, all-inclusive cluster
    - At each step, split a cluster until each cluster contains a point (or there are k clusters)
- Traditional hierarchical algorithms use a similarity or distance matrix
  - Merge or split one cluster at a time

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Agglomerative Clustering Algorithm

- More popular hierarchical clustering technique
- Basic algorithm is straightforward
  1. Compute the proximity matrix
  2. Let each data point be a cluster
  3. **Repeat**
  4. Merge the two closest clusters
  5. Update the proximity matrix
  6. **Until** only a single cluster remains
- Key operation is the computation of the proximity of two clusters
  - Different approaches to defining the distance between clusters distinguish the different algorithms

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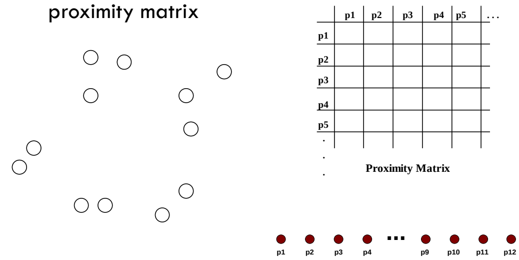
Starting Situation

- Start with clusters of individual points and a proximity matrix



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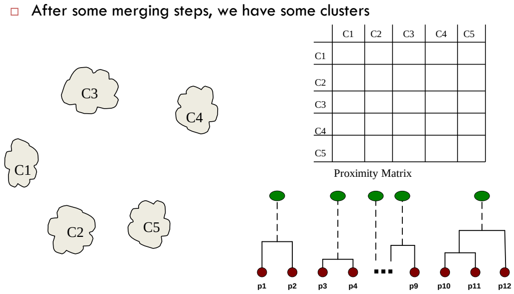
Intermediate Situation

- After some merging steps, we have some clusters



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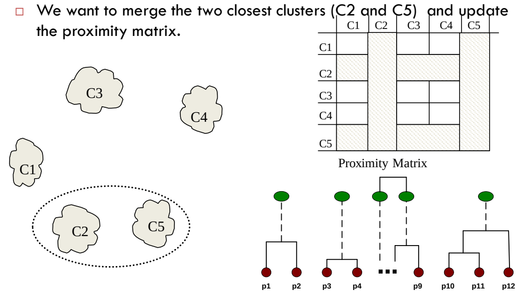
Intermediate Situation

- We want to merge the two closest clusters (C2 and C5) and update the proximity matrix.



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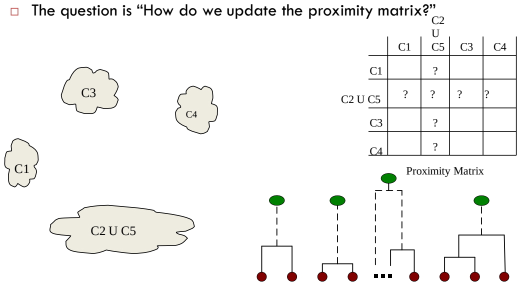
After Merging

- The question is "How do we update the proximity matrix?"



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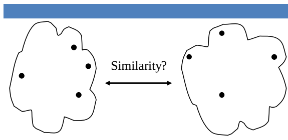
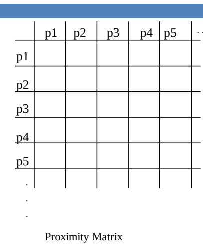
How to Define Inter-Cluster Similarity





- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
  - Ward's Method uses squared error

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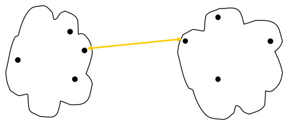
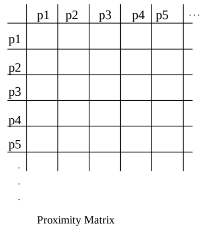
How to Define Inter-Cluster Similarity





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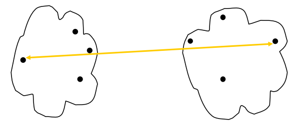
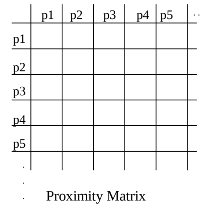
How to Define Inter-Cluster Similarity





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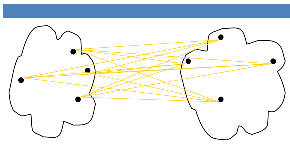
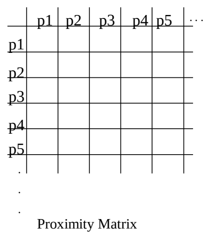
How to Define Inter-Cluster Similarity





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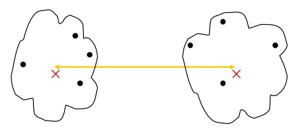
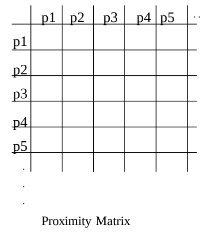
How to Define Inter-Cluster Similarity





- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
  - Ward's Method uses squared error

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Distance measures

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Euclidiean distance (Statistical Distance)

d(p,q) = $\sqrt{(q1-p1)^2+(q2-p2)^2+\cdots+(qn-pn)^2}$

= $\sqrt{\sum_{i=1}^{n}(qi-pi)^2}$

- Manhattan distance

d(x,y) = $\sum_{i=1}^{n}|xi-yi|$

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Distance measures

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Given two objects represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8):

(a) Compute the *Euclidean distance* between the two objects.

(b) Compute the *Manhattan distance* between the two objects.

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Distance measures

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Given two objects represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8):

(a) Compute the *Euclidean distance* between the two objects.

(b) Compute the *Manhattan distance* between the two objects.

Answer:

(a) Compute the *Euclidean distance* between the two objects.

$$\begin{aligned} d(i,j) &= \sqrt{(x_{i1}-x_{j1})^2+(x_{i2}-x_{j2})^2+\cdots+(x_{in}-x_{jn})^2} \\ &= \sqrt{|22-20|^2+|1-0|^2+|42-36|^2+|10-8|^2} = 6.71 \end{aligned}$$

(b) Compute the *Manhattan distance* between the two objects.

$$\begin{aligned} d(i,j) &= |x_{i1}-x_{j1}|+|x_{i2}-x_{j2}|+\cdots+|x_{in}-x_{jn}| \\ &= |22-20|+|1-0|+|42-36|+|10-8| = 11 \end{aligned}$$

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Cluster Similarity: MIN or Single Link

- Similarity of two clusters is based on the two most similar (closest) points in the different clusters
  - Determined by one pair of points, i.e., by one link in the proximity graph.

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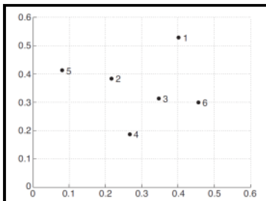

Point	x Coordinate	y Coordinate
p1	0.40	0.53
p2	0.22	0.38
p3	0.35	0.32
p4	0.26	0.19
p5	0.08	0.41
p6	0.45	0.30

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

Euclidean distance matrix for 6 points

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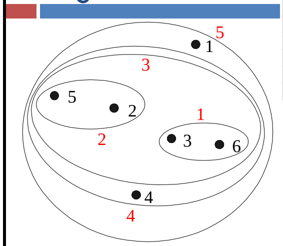
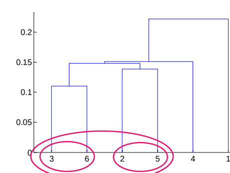
Hierarchical Clustering: MIN or Single Link

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00



Nested Clusters



Dendrogram

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Strength of MIN

Original Points

Two Clusters

- Can handle non-elliptical shapes

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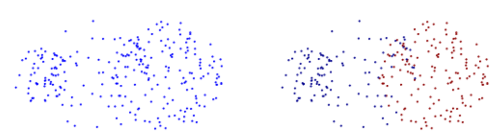
Limitations of MIN



Original Points

Two Clusters

- Sensitive to noise and outliers

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Cluster Similarity: MAX or Complete Linkage

- Similarity of two clusters is based on the two least similar (most distant) points in the different clusters
  - Determined by all pairs of points in the two clusters

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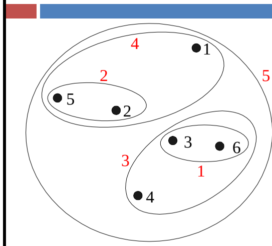
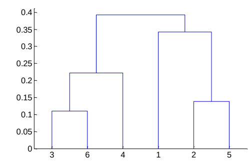
Hierarchical Clustering: MAX

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00



Nested Clusters



Dendrogram

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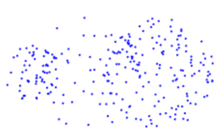
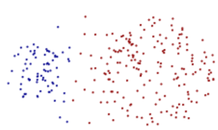
Strength of MAX



Original Points



Two Clusters

- Less susceptible to noise and outliers

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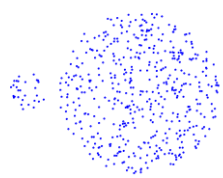
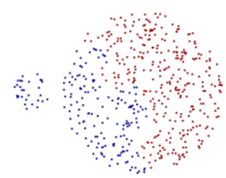
Limitations of MAX



Original Points



Two Clusters

- Tends to break large clusters
- Biased towards globular clusters

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Cluster Similarity: Group Average

- Proximity of two clusters is the average of pairwise proximity between points in the two clusters.

$$\text{proximity}(\text{Cluster}_i, \text{Cluster}_j) = \frac{\sum\limits_{\substack{p_i \in \text{Cluster}_i \\ p_j \in \text{Cluster}_j}} \text{proximity}(p_i, p_j)}{|\text{Cluster}_i| * |\text{Cluster}_j|}$$

- Need to use average connectivity for scalability since total proximity favors large clusters

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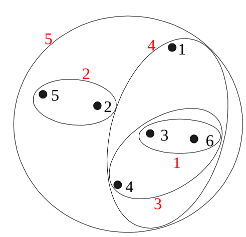
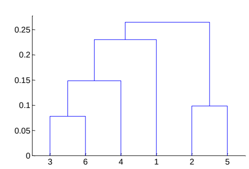
Hierarchical Clustering: Group Average



Nested Clusters



Dendrogram

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Hierarchical Clustering: Group Average

- Compromise between Single and Complete Link
- Strengths
  - Less susceptible to noise and outliers
- Limitations
  - Biased towards globular clusters

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Cluster Similarity: Ward's Method

- Similarity of two clusters is based on the increase in squared error when two clusters are merged
  - Similar to group average if distance between points is distance squared
- Less susceptible to noise and outliers
- Biased towards globular clusters
- Hierarchical analogue of K-means
  - Can be used to initialize K-means

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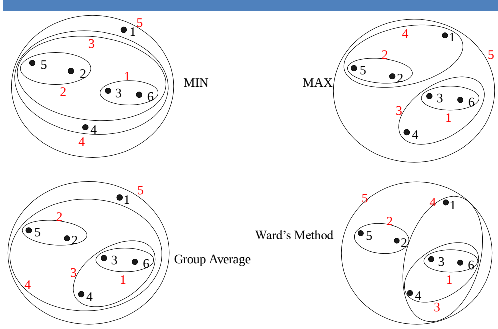
Hierarchical Clustering: Comparison



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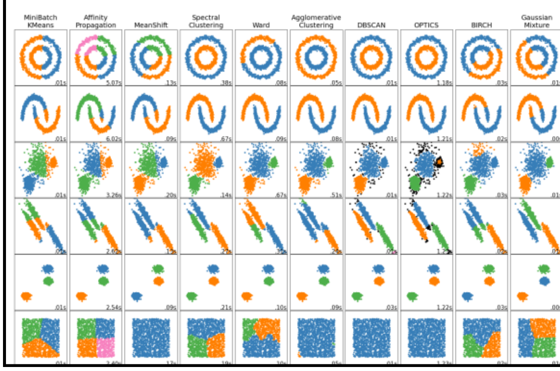
Overview of clustering methods



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Reading materials

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- https://scikit-learn.org/stable/modules/clustering.html

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Examples

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- Example 1
  - Agglomerative Clustering Algorithm, Single linkage, Euclidean distance
- Example 2
  - Agglomerative Clustering Algorithm, Complete linkage, Euclidean distance

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